GS1 DataMatrix UDI Label Management System — Master Build Instructions

For Claude Code: Read ALL files in this directory before writing any code.

File Reading Order

1. `MASTER-INSTRUCTIONS.md` (this file) — overview, architecture, product catalog
2. `AGENT-1-foundation.md` — Phase 1–3: project setup, auth, barcode engine, data form
3. `AGENT-2-designer.md` — Phase 4–5: template designer, ZPL generation, export
4. `AGENT-3-printing.md` — Phase 6–8: printer module, batch printing, deployment
5. `ui-preview.tsx` — reference UI implementation (React component with real GS1 encoding)

Build phases sequentially. Test each phase before moving to the next.

What This App Does

A web application for creating, previewing, and printing GS1 DataMatrix UDI labels for medical devices. Complies with FDA 21 CFR 801.45 and EU MDR 2017/745. Runs on a local PC with a Zebra GK420D USB printer, accessible to up to 6 users on the same office LAN.

Architecture

- **Frontend:** React 18 + TypeScript + Vite + Tailwind CSS
- **Backend:** Node.js + Express + TypeScript
- **Storage:** JSON files in `/data/` directory (NO SQL database — max 6 users)
- **Barcode:** bwip-js (gs1datamatrix symbology)
- **Printer:** Zebra Browser Print (USB), raw TCP (network)
- **Auth:** JWT in HTTP-only cookies, bcrypt passwords
- **Validation:** Zod (backend), react-hook-form + zod (frontend)
- **File locking:** proper-lockfile for concurrent JSON access

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THE #1 BUSINESS RULE

LOT MODE: (01) GTIN + (10) Lot + (11) Manufacturing Date. **NO expiration date anywhere. SERIAL MODE:** (01) GTIN + (17) Expiration + (11) Manufacturing + (21) Serial. **WITH expiration date everywhere.**

This rule is enforced at every layer: form fields, GS1 encoding, DataMatrix barcode, HRI text, ZPL output, PDF/PNG export, and printed label. Violation is a regulatory compliance failure.

HRI Display Order (human-readable text on label):

- LOT: (01) GTIN → (10) LOT → (11) MfgDate
- SERIAL: (01) GTIN → (21) SERIAL → (11) MfgDate → (17) ExpDate

DataMatrix Encoding Order (inside barcode — different from HRI):

- LOT: (01)GTIN(11)YYMMDD(10)LOT
- SERIAL: (01)GTIN(17)YYMMDD(11)YYMMDD(21)SERIAL

These two orders are intentionally different. Encoding puts fixed-length AIs first. HRI puts the identifier right after GTIN for readability.

Complete Product Catalog (39 products)

All 39 products are pre-seeded on first run. Users select a product — the mode, GTIN, lot prefix, serial prefix, and shelf life are all determined by the product selection. No manual mode toggle.

GTIN Note

All GTINs are 14 digits with 00 prefix. The source data had leading zeros stripped by Excel. Every GTIN in this document is the correct 14-digit GTIN-14.

Product Name Format

{PartNumber} - {DeviceName} (e.g., PV2010E - BWIII Basics)

SERIAL MODE — 9 products (with 7-year expiration)

BWIII Family — serial prefix: BWIII{YYYY}-, start at 3000

Product Name	GTIN-14	Lot Prefix	Serial Prefix
PV2010E - BWIII Basics	00850008393174	—	BWIII
PV2006E - BWIII EEG	00850008393181	—	BWIII
PV2007E - BWIII EEG Plus	00850008393198	—	BWIII
PV2008E - BWIII PSG	00850008393204	—	BWIII
PV2009E - BWIII PSG Plus	00850008393211	—	BWIII

BWMini Family — serial prefix: `BWM{YYYY}-`, start at 3000

Product Name	GTIN-14	Lot Prefix	Serial Prefix
PV2310 - BWMini EEG	00850008393235	—	BWM
PV2312 - BWMini HST	00850008393242	—	BWM
SI1842 - BWMini HST Compass	00850008393259	—	BWM
PV2311 - BWMini PSG	00850008393228	—	BWM

Serial Number Rules:

- Serials are globally unique within each prefix family
- BWIII2026-3000 could be ANY BWIII model — one shared counter per prefix+year
- BWM2026-3000 could be ANY BWMini model — separate counter from BWIII
- System tracks last-used serial per prefix family per year in `serial-counters.json`
- Next batch auto-suggests `lastUsed + 1`; user can adjust

LOT MODE — 30 products (no expiration)

Lot number format: `{lotPrefix}-{YYMMDD}` where YYMMDD = manufacturing date. Lot number is auto-generated when user picks product + manufacturing date. User can override. One lot covers all units of that product manufactured on that date (could be hundreds).

Maxxi Gold Electrodes (10)

Product Name	GTIN-14	Lot Prefix
PV1010-33CI - Maxxi Gold Electrode 48"	00850008393044	MG48CI
PV1010-13CI - Maxxi Gold Electrode 60"	00867975000295	MG60CI
PV1010-23CI - Maxxi Gold Electrode 96"	00850008393006	MG96CI
SI1872 - Maxxi Gold Snap Electrode 48"	00850008393037	MS48CI
SI1873 - Maxxi Gold Snap Electrode 60"	00850008393020	MS60CI
SI1778B - Maxxi Gold Snap Electrode 96"	00850008393013	MS96CI
SI2036 - Maxxi Gold AgCl Electrode 60" Disposable	00850008393303	MA60CI
SI2137 - Maxxi Gold AgCl Electrode 60" Reusable	00850008393358	MAR60CI
SI2037 - Maxxi Gold AgCl Electrode 96" Disposable	00850008393365	MA96CI
SI2138 - Maxxi Gold AgCl Electrode 96" Reusable	00850008393310	MAR96CI

Maxxi Flow Sensors (3)

Product Name	GTIN-14	Lot Prefix
SI1775B - Maxxi Flow Sensor 3ft Key Connector	00850008393341	MF3KCI
SI1776B - Maxxi Flow Sensor 7ft Key Connector	00850008393334	MF7KCI
SI1486C - Maxxi Flow Sensor 7ft TP Connector	00850008393327	MF7SCI

Maxxi Position Sensors (3 — shared GTIN 00850008393297)

Product Name	GTIN-14	Lot Prefix
SI1247U - Maxxi Position Sensor AC 1.0	00850008393297	MP
SI1247A - Maxxi Position Sensor AC 1.1	00850008393297	MPAC
SI1247D - Maxxi Position Sensor DC	00850008393297	MPDC

These three share one GTIN — they are the same product with different compatibility. System allows duplicate GTINs across products.

Maxxi RIP Respiratory (10)

Product Name	GTIN-14	Lot Prefix
SI2121 - Maxxi RIP Abdomen Interface 3ft	00850008393136	IA3
SI2124 - Maxxi RIP Abdomen Interface 7ft	00850008393143	IA7
SI1659 - Maxxi RIP Adjustable Inductive Belt	00850008393105	AB
SI1660 - Maxxi RIP Infant Inductive Belt	00850008393051	IB
SI1693 - Maxxi RIP Large Inductive Belt	00850008393075	LB
SI1665 - Maxxi RIP Pediatric Inductive Belt	00850008393068	PB
SI2125 - Maxxi RIP Thorax Interface 3ft	00850008393129	IT3
SI2123 - Maxxi RIP Thorax Interface 7ft	00850008393112	IT7
SI1694 - Maxxi RIP X-Large Inductive Belt	00850008393082	XLB
SI1662 - Maxxi RIP XX-Large Inductive Belt	00850008393099	XXLB

Maxxi Snore Sensors (3)

Product Name	GTIN-14	Lot Prefix
SI1866B - Maxxi Snore 2ft Key Connector	00850008393266	MS2K
SI1784B - Maxxi Snore 7ft Key Connector	00850008393273	MS7K
SI1487B - Maxxi Snore 7ft TP Connector	00850008393280	MS7S

Maxxi Cannula (3)

Product Name	GTIN-14	Lot Prefix
SI1058 - Maxxi Cannula 7ft Adult Nasal/Oral	00850008393389	MCANO7
SI1057 - Maxxi Cannula 7ft Nasal	00850008393396	MCAN7
SI1195 - Maxxi Cannula 2ft Nasal	00850008393372	MCAN2

Data Storage Files

```
/data/
├── users.json            # User accounts (max 6)
├── products.json         # Product profiles (39 pre-seeded)
├── templates.json        # Label templates (all versions)
├── printers.json         # Configured printers
├── print-jobs.json       # Print job history
├── serial-counters.json  # Last-used serial per prefix family per year
└── audit-log.json        # Audit trail (append-only)
```

Product Schema

```json
{
  "id": "uuid",
  "name": "PV1010-23CI - Maxxi Gold Electrode 96\"",
  "gtin": "00850008393006",
  "partNumber": "PV1010-23CI",
  "description": "",
  "identifierMode": "lot",
  "lotPrefix": "MG96CI",
  "serialPrefixFamily": null,
  "shelfLifeYears": 0,
  "shelfLifeMonths": 0,
  "shelfLifeDays": 0,
  "autoExpirationEnabled": false,
  "createdBy": "system",
  "createdAt": "2026-02-18T00:00:00Z",
  "updatedAt": "2026-02-18T00:00:00Z"
}
```

Serial Counter Schema

```json
[
  {
    "id": "bwiii-2026",
    "prefixFamily": "BWIII",
    "year": 2026,
    "lastUsedNumber": 3047,
    "updatedAt": "2026-02-18T10:30:00Z"
  },
  {
    "id": "bwm-2026",
    "prefixFamily": "BWM",
    "year": 2026,
    "lastUsedNumber": 3012,
    "updatedAt": "2026-02-18T10:30:00Z"
  }
]
```

Key UX Decisions

1. **Product selector drives everything** — user picks a product, and mode/GTIN/prefixes auto-fill. No manual lot/serial toggle.
2. **Lot numbers auto-generate** — `{lotPrefix}-{YYMMDD}` from product + manufacturing date. Editable for override.
3. **Serial numbers auto-suggest** — reads last-used counter, suggests next range. Both start and end are editable.
4. **Default template** — Square 1"×1" on first load.
5. **HRI text alignment** — left/center/right control on the preview and stored in template.
6. **Label content scale** — slider (50%–200%) to adjust DataMatrix + text size within the label.
7. **Square label layout** — DataMatrix on top, text below (stacked). Wide labels use side-by-side layout.
8. **Export formats** — PDF (label-sized pages), PNG, ZPL. Batch export up to 500 labels.
9. **LAN printing** — LAN users queue jobs; printer PC operator clicks "Print Now".
10. **Duplicate GTINs allowed** — Maxxi Position variants share one GTIN.